

Amin Skenderi

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EDUCATION

Technical University Berlin

Expected Graduation: 2027

- B.Sc. Computer Engineering, 4th semester (on track to complete in 5 semesters)
- Relevant Coursework: Data Structures & Algorithms (Top 5%), Theoretical Computer Science (top score in class), Semiconductor Physics for Electrical Engineers (Top 10%), Signals and Systems (Top 5%)

Hannah-Arendt-Gymnasium

Graduation Year: 2024

- Computer Science Award (“200€ for excellent performance”)
- Final grade: 1.2 (German system); focus on Computer Science, Physics and Math

EXPERIENCE

Software Engineering Intern (C++) | Cybus GmbH

01.2026 – 06.2026

- Developed and maintained **C++** components for the Connectware industrial IoT platform, enabling data exchange between shop-floor machines and cloud services
- Implemented protocol connectors for **OPC UA**, **MQTT** and **Modbus/TCP**, normalizing heterogeneous device data into a unified internal data model
- Optimized message processing in a multi-threaded data pipeline, reducing end-to-end latency for high-frequency sensor streams
- Collaborated in an agile team using **Git**, code reviews and Jira, contributing to architecture discussions for new device drivers

Coding Trainer | CodingGiants GmbH

08.2023 – 01.2026

- Instructed teenagers in **Python**, covering advanced concepts such as algorithms and structured programming
- Taught programming fundamentals to children using **Scratch**, focusing on logical thinking and problem-solving

PROJECTS

LLM Document Analysis Pipeline | github.com/AmkoAmin/llm-doc-pipeline

2026

- Built a **Python**-based **FastAPI** backend that answers natural-language questions over uploaded PDFs, by implementing a **RAG** pipeline with **LangChain**, **Claude** (Sonnet 4.6), **Voyage** embeddings, and a **FAISS** index, containerized with **Docker**
- Hardened the service for public deployment, by adding domain-scoped CORS, per-IP rate limiting, upload-size limits, and authentication fixes
- Validated correctness and reliability with an automated unit and integration test suite using **pytest**
- Designed a self-hosted deployment architecture, by serving a static **GitHub Pages** frontend through a **Cloudflare Tunnel** to the Dockerized API

Autonomous Robot with ROS 2 | github.com/AmkoAmin/ros2-autonomous-robot

2026

- Led a 3-person team building an autonomous robot with **C++** using **ROS 2** on a **Raspberry Pi** that localizes itself in a maze and navigates to arbitrary target cells
- Implemented grid-based localization, by detecting maze walls from **Lidar** data via Hough transform and matching observations against a known map, validated with a custom test harness
- Developed **Dijkstra** path planning and precise closed-loop motion control, by combining wheel-encoder odometry with P-controllers, within a modular **ROS 2** node architecture

Distributed HTTP Server & Chord DHT | github.com/AmkoAmin/chord-dht-server

2025

- Served thousands of concurrent connections on a single node, by building a custom HTTP server in **C** with low-level socket programming, manual request parsing, and memory safety verification using **Valgrind**
- Enabled scalable resource discovery across distributed nodes, by implementing a **Chord DHT** with consistent hashing and finger table routing
- Delivered fault-tolerant task distribution, by developing a **ZeroMQ**-based MapReduce-style system with retry logic

SKILLS

Programming/Scripting Languages: C, C++, Python, Java, MATLAB, MIPS-Assembler, SQL, Bash
Tools: Linux, Git, GitHub (CI/CD), Docker (Compose), CMake, colcon, Azure, MySQL, Arduino, Blender
Frameworks & Technologies: FastAPI, Flask, ROS 2, MQTT, Valgrind, ZeroMQ, REST
AI & LLM: OpenAI API, Anthropic API, LangChain/LangGraph, RAG pipelines, Agentic Workflows
Languages: German (Native/C2), English (Fluent/C1)